Original Investigation

Prophylactic Use of a Probiotic in the Prevention of Colic, Regurgitation, and Functional Constipation
A Randomized Clinical Trial

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IMPORTANCE Infantile colic, gastroesophageal reflux, and constipation are the most common functional gastrointestinal disorders that lead to referral to a pediatrician during the first 6 months of life and are often responsible for hospitalization, feeding changes, use of drugs, parental anxiety, and loss of parental working days with relevant social consequences.

OBJECTIVE To investigate whether oral supplementation with *Lactobacillus reuteri* DSM 17938 during the first 3 months of life can reduce the onset of colic, gastroesophageal reflux, and constipation in term newborns and thereby reduce the socioeconomic impact of these conditions.

DESIGN A prospective, multicenter, double-masked, placebo-controlled randomized clinical trial was performed on term newborns (age <1 week) born at 9 different neonatal units in Italy between September 1, 2010, and October 30, 2012.

SETTING Parents were asked to record in a structured diary the number of episodes of regurgitation, duration of inconsolable crying (minutes per day), number of evacuations per day, number of visits to pediatricians, feeding changes, hospitalizations, visits to a pediatric emergency department for a perceived health emergency, pharmacologic interventions, and loss of parental working days.

PARTICIPANTS In total, 589 infants were randomly allocated to receive *L reuteri* DSM 17938 or placebo daily for 90 days.

INTERVENTIONS Prophylactic use of probiotic.

MAIN OUTCOMES AND MEASURES Reduction of daily crying time, regurgitation, and constipation during the first 3 months of life. Cost-benefit analysis of the probiotic supplementation.

RESULTS At 3 months of age, the mean duration of crying time (38 vs 71 minutes; $P < .01$), the mean number of regurgitations per day (2.9 vs 4.6; $P < .01$), and the mean number of evacuations per day (4.2 vs 3.6; $P < .01$) for the *L reuteri* DSM 17938 and placebo groups, respectively, were significantly different. The use of *L reuteri* DSM 17938 resulted in an estimated mean savings per patient of €88 (US $118.71) for the family and an additional €104 (US $140.30) for the community.

CONCLUSIONS AND RELEVANCE Prophylactic use of *L reuteri* DSM 17938 during the first 3 months of life reduced the onset of functional gastrointestinal disorders and reduced private and public costs for the management of this condition.

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